

Fernando Román Hidalgo

ROBOTICS AND AUTOMATION ENGINEER

41909, Salteras, Sevilla, Spain

☎ 611467316 ✉ ferromhid@gmail.com 🗣 FerXxx 🌐 Fernando Román Hidalgo 🏠 Portfolio



Summary

Engineer with a **multidisciplinary profile and systemic vision**, featuring professional experience in developing technological solutions and currently completing postgraduate studies. I possess a **high capacity for abstraction in solving complex problems** and **accelerated self-taught learning**, enabling me to provide significant value in technically demanding projects. My objective is to consolidate an ambitious career plan, with a focus on automation and embedded software, within dynamic and innovative environments where continuous improvement is the standard.

Experience

Software Engineer

 Artificial Aerospace & Defense | Seville, Spain

FULLSTACK DEVELOPMENT, DEVOPS, AUTOMATION AND PROJECT MANAGEMENT

April 2025 - May 2026

- **Management and Client Relations:**
 - Project management for the development of a process automation application for a major aerospace company.
 - Direct customer interaction for requirements definition and specification.
 - Technical leadership and coordination in the development of custom solutions.
 - Application of the Agile Scrum framework integrated with Lean principles to optimize workflow and eliminate inefficiencies in development.
- **Technical Development and Automation:**
 - Use of Git for version control in both projects.
- **Web Application:**
 - Development of a web application with Google App Script (JavaScript, HTML, CSS) to replace obsolete Excel spreadsheets with an intelligent and automated interface.
- **Digital Twin and Optimization Platform:**
 - Design and implementation of a digital twin for a composite material production plant, with a FastAPI and PostgreSQL backend, React frontend, deployed using Docker containers.
 - Production deployment: environment configuration, port opening, Nginx reverse proxy for traffic management, and full system commissioning.
 - Development of predictive AI models in Python for early failure detection using industrial sensor data.
 - Implementation of complex production scheduling algorithms in Python using Google OR-Tools' CP-SAT solver, with variable capacity-load constraints and custom priority weighting systems.
 - Orchestration of data pipelines and execution of AI models with Apache Airflow.
 - Integration with SAP for synchronizing master data and production planning transactions.
 - Acquisition and interpretation of data from industrial sensors to feed the digital twin and predictive models.

Projects

To see all my projects in detail, check out my **Interactive Portfolio**.

Honors and Awards

Winner

MINISUMO CATEGORY

National Competition Robotics League (LNRC) | Spain

2024

Education

Master's Degree in Electronic Engineering, Robotics and Automation	 University of Seville Seville, Spain 2025 - 2026
Bachelor's Degree in Electronic Engineering, Robotics and Mechatronics GPA: 7.5/10	 University of Seville Seville, Spain 2021 - 2025
Technological Baccalaureate (Graduated with Honors distinction)	 IES Heliche Seville, Spain 2019 - 2021

Certifications

Google Developer Program — Machine Learning COMPLETED MACHINE LEARNING COURSES WITHIN THE GOOGLE DEVELOPER PROGRAM.	 Google Developers 2026
AI with Google Gemini TRAINING PROGRAM ON DEVELOPING AND MODELING SOLUTIONS USING GOOGLE GEMINI GENERATIVE AI.	Banco Santander & Google 2026
GitHub Copilot — Software Development PRACTICAL COURSE ON AI-ASSISTED SOFTWARE DEVELOPMENT AND WORKFLOW OPTIMIZATION.	GitHub 2026
Advanced Python Course ONLINE	 Idexa Jan 2026
Certificate in Advanced English (C1 Advanced)	 Cambridge Assessment English Sep 2024
Basic Life Support and Automated External Defibrillation	 Ympetu 2020
Aquatic Lifeguard	 Ympetu 2020

Skills

Programming	C, C++, Python (OpenCV), MATLAB, Simulink, ROS/ROS2, JavaScript, React, VBA, HTML, CSS, SQL, SAP/HANA
Electronics	Digital/Analog electronics, microcontrollers (MSP430, ESP32, Arduino...), RTOS, FPGA, automation and control
Simulation	System modeling and simulation (Gazebo, Webots), PID control design, embedded systems integration
DevOps	Docker, Kubernetes, AWS, Linux, Git, CI/CD, Apache Airflow
AI / LLMs	Language Model APIs (Gemini, Claude, Copilot...), Prompt Engineering, Machine Learning applied (LSTM, predictive models), Reinforcement Learning (DRL)
Protocols	TCP/IP, MQTT, Zigbee, CAN bus, UART, I2C, SPI
3D Design	Fusion 360, FreeCAD, KiCad

Soft Skills

Capabilities

Communication, leadership, adaptation, proactivity, critical thinking, punctuality, autonomy, management and teamwork

Languages

Spanish

Native

English

Professional competence (C1)

French

Basic competence